

Aaron Bullock

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EDUCATION

Duke University

Software Engineering Track, M.Eng. in Electrical & Computer Engineering; GPA: 3.6

Expected Graduation: May 2027

Durham, NC

Duke University

BSE in Electrical & Computer Engineering, Computer Science; GPA: 3.7

Aug 2022 – May 2026

Durham, NC

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Impulse Space

Los Angeles, CA

- Architected and developed a company-wide platform for visualizing, configuring, and managing aerospace vehicle architectures, reducing production engineering workflow time by 86%, from 14 hours to 2 hours.
- Built an interactive graph-based interface with TypeScript, React, and React Flow and designed a PostgreSQL-backed service layer using Prisma.
- Containerized and deployed the application with Docker, Next.js, and Nginx, using GitLab CI/CD and webhooks to automate build and deployment workflows.
- Developed Python integration pipelines that ingest and transform engineering data from CAD systems and internal APIs.

Embedded Software Engineer Intern

May 2025 – Aug 2025

Persistent Systems, LLC

New York, NY

- Developed a Gradle-packaged Java Quality Control application that integrated C device firmware with Amazon RDS for SQL Server through JDBC, removing a production bottleneck and enabling military-device shipments.
- Standardized error-code reporting across all device QC applications and extended Amazon RDS for SQL Server to store validation metrics from the new device.
- Implemented asynchronous device validation with Java SwingWorker, preventing UI freezes and displaying real-time status.
- Reduced validation time for a new device QC application by 50%, from 12 to 6 minutes, through workflow optimization.
- Expanded unit-test coverage and standardized application builds with Bash scripts, reducing build time by 50% while preventing database configuration errors.

Computer Engineering Intern

May 2024 – Aug 2024

Federal Aviation Administration

Atlantic City, NJ

- Led development of simulations for an FAA evaluation platform, reducing test execution time by 66%.
- Increased data-processing throughput to 5x its previous rate by implementing multithreading and parallel execution using C++ and POSIX threads.
- Developed and deployed Python automation scripts integrating FAA APIs, reducing required in-lab testing time by 50%.

PROJECTS

Trybl | *Python, Go, PostgreSQL, Docker, FastAPI, GPT-4o*

- Built the discovery service for Trybl, a social activity platform that generates and ranks personalized activity recommendations based on users' location, availability, preferences, budget, and interaction history.
- Developed the service with Python and FastAPI, using GPT-4o to generate context-aware activities and PostgreSQL to cache recommendations by community.
- Implemented semantic deduplication with Model2Vec embeddings, preventing users from repeatedly seeing similar events.
- Integrated the discovery service with Trybl's Go backend using authenticated APIs, batch processing, concurrency locks, timeouts, retries, and exponential backoff.

Publisher Accounting System | *TypeScript, Next.js, Tailwind CSS, React, Supabase, Prisma, PostgreSQL, Docker*

- Architected and developed a production full-stack publishing platform that replaced spreadsheet workflows for managing books, sales, royalties, author payments, and publisher profit.
- Designed the PostgreSQL data model and Prisma migration strategy, enforcing relational integrity and supporting multi-currency transactions, multiple sales channels, and royalty calculations.
- Built validated ingestion pipelines for Amazon, IngramSpark, and BackerKit sales reports, transforming XLSX and CSV data into normalized transactions and generating financial reports through server-side APIs.
- Created Dockerized QA and production environments with Nginx and GitHub Actions CI/CD, integrating Supabase Auth and Storage with health checks and verified database and media backups.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C/C++, HTML/CSS, Bash, Go

Frontend & Backend: React, Next.js, Node.js, FastAPI, Django, Express, Tailwind CSS

Data & Infrastructure: PostgreSQL, SQL Server, JDBC, Supabase, Amazon RDS, Docker, GitLab CI/CD, GitHub Actions